Hassaan Raza

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Professional Summary

Final year Software Engineering student specialising in AI/ML engineering: LLMs, agentic pipelines, and RAG systems. Independently designed, built, and shipped 10+ AI applications, from a legal-intelligence RAG tool to a transformer built from scratch.

Technical Skills

LLMs & Agents: LangChain, LangGraph, CrewAI, Agno, RAG, ChromaDB, HuggingFace Transformers, LoRA/QLoRA

ML & Deployment: PyTorch, TensorFlow, scikit-learn, Streamlit, FastAPI, Google Cloud Run, Docker

APIs & Models: Ollama Cloud, OpenRouter, Gemini, OpenAI, Groq, MCP Servers

Data & CV: Power BI, Tableau, PostgreSQL, MySQL, OpenCV, PyMuPDF

Languages: Python, SQL, JavaScript (basic)

Projects

VerdictAI | *RAG · ChromaDB · HuggingFace · Ollama Cloud · PyMuPDF*

- Built a custom RAG pipeline from scratch (PyMuPDF extraction, ChromaDB vector store, Gemma 4 31B via Ollama Cloud) to close the gap between complex legal contracts and the non-lawyers who sign them; scores risk 1-10 with clause-level citations, live and publicly testable.

NanoCodeGPT | *PyTorch · GPT-2 BPE Tokenizer · Google Colab*

- Built a decoder-only GPT transformer from scratch in PyTorch, no pretrained weights, to prove out transformer internals beyond calling an API; trained on 8,000 Python functions on a free Colab T4 GPU, dropping train loss from 10.9 to 0.13 in under an hour.

Jobee | *Streamlit · Sentence Transformers · Multi-API Integration*

- Built an AI job-matching platform to fix the hundreds-of-irrelevant-listings problem on standard job boards; parses a candidate's CV, queries 7 job board APIs in parallel, and ranks results by semantic fit and a 6-tier geo-proximity score instead of keyword match.

MediScan | *Gemini Vision · Multimodal AI · Streamlit*

- Co-built a multimodal clinical diagnostic suite to support earlier patient self-triage; Gemini Vision generates structured diagnoses with severity/confidence scores from X-rays, MRIs, CT scans, and reports. Presented at a Google developer event in Malaysia.

ERP Chatbot & Sales Assistant | *Client project for Eccountant · CrewAI · MySQL · AWS RDS*

- Delivered freelance client work for Eccountant, a live business, to replace manual multi-source reporting; built a multi-company ERP AI assistant with three specialised agents (sales, inventory, cash flow) on write-protected database access, plus a natural-language sales assistant for the CEO.

WayFairy | *CrewAI · Agentic AI Travel Planner*

- Built a four-agent CrewAI travel planner to replace manual multi-tab trip research; agents handle logistics, activities, packing, and itinerary synthesis using live web search rather than static training data, with a full budget breakdown.

Education

BS Software Engineering | University of Management and Technology (UMT), Lahore | 2023 - 2027 (Expected)

Relevant coursework: Software Quality Engineering, Information Security, Deep Learning, Computer Vision. Completed Google Cloud AISeekho challenge 2025 and 2026, deploying to Cloud Run.

Certifications

Engineer AI Agents with ADK · Gen AI Apps with Gemini & Streamlit · Intro to LLMs — Google Cloud